

Researchers studying the metabolic profile of HeLa cells collected the cells in buffer, gently lysed the plasma membrane, and then centrifuged the lysate to perform subcellular fractionation and remove intact organelles. The supernatant was subjected to liquid chromatography–mass spectrometry to measure the concentration of cytosolic metabolites. Which metabolite is LEAST likely to be found in the lysate supernatant?

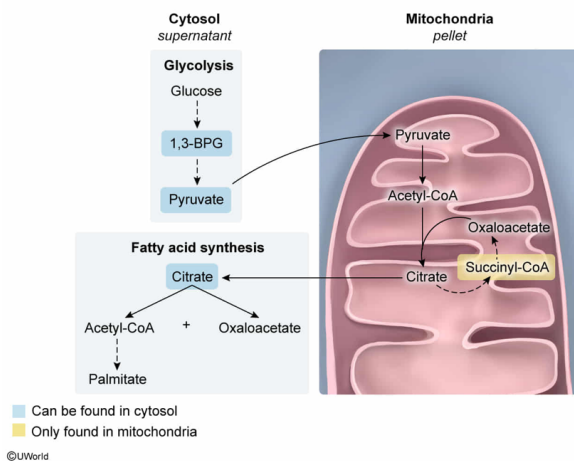
- ☐ A. Citrate [17%]
- ☐ B. Pyruvate [8%]
- ☐ C. 1,3-Bisphosphoglycerate [25%]
- ✔ ☒ D. Succinyl-CoA [47%]

Correct

47% answered correctly

## Explanation:

Some metabolites can move compartments, but succinyl-CoA remains in the mitochondria



**Subcellular fractionation**—the experimental separation of a cell lysate by cellular compartment—followed by **liquid chromatography–mass spectrometry** (LC-MS) metabolomic screening is an effective means of identifying and measuring metabolites in different **cellular compartments**. After fractionation, **cytosolic metabolites are found in the supernatant** whereas metabolites within **organelles** (eg, **mitochondria**) are retained in the organelles as they **sink to the pellet**.

**Glycolysis** is a metabolic pathway that occurs in the cytosol; therefore, intermediates within this pathway (eg, **1,3-bisphosphoglycerate**) as well as the product of the pathway (ie, **pyruvate**) are expected to be present at some level within the cytosolic supernatant (**Choices B and C**). Pyruvate is also

expected to be in the mitochondria because some of it is transported to the mitochondria after glycolysis for further catabolism.

Within the mitochondria, pyruvate is oxidatively decarboxylated to form acetyl-CoA, which then enters the **citric acid cycle**. Although the citric acid cycle is a mitochondrial matrix pathway, some citric acid cycle metabolites can leave the mitochondria and go back into the cytosol. For example, **citrate** is transported from the mitochondrial matrix **to the cytosol**, where it is cleaved to form cytosolic acetyl-CoA necessary for **fatty acid synthesis**. Therefore, at least some citrate is likely to be found in the supernatant (**Choice A**).

However, the question asks for the metabolite that is *least* likely to be in the supernatant. Succinyl-CoA is synthesized as an intermediate of the citric acid cycle and as a product of **odd-chain fatty acid catabolism**, both of which occur in the **mitochondrial matrix**. Succinyl-CoA is *not* transported out of the mitochondria into the cytosol. Therefore, of the available choices **succinyl-CoA is least likely to be found in the supernatant**, which represents the cytosolic fraction.

### **Educational objective:**

Metabolic intermediates (ie, metabolites) can be either segregated to one cellular compartment or may be found in multiple cellular compartments, depending on the metabolic pathways in which a particular metabolite is involved.

**Time spent:**0 sec

**Subject:**  
Biochemistry

**QID:**403788

**System:**  
Metabolic Reactions